SRS Report

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	15
Clarity	3					
Content Quality	6					
Spelling & Grammar	3					
Organization and Formatting	3					
Total obtained mark						
Comments						

Semester: Fall……2026……

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Batch: 42 **Section: A1** **Course Code: SE-331**

Course Name: Software Engineering Design Capstone Project

Course Teacher Name: Rahat Uddin Azad **Designation: Lecturer**

Submission Date: …15…. /…10…/…2026….

FabricIQ AI

AI-Powered Fabric Defect Inspection and Quality Analysis System

1. Introduction

FabricIQ AI is a web-based fabric inspection system designed to assist users in identifying selected visible fabric defects using artificial intelligence. The system will allow users to upload fabric images or capture images through a supported camera. It will analyze the submitted image, display predicted defects, estimate dominant colors, and provide a preliminary visual inspection score.

The system will also support human review, inspection history, and report generation. Its main AI feature will be fabric defect detection using a YOLO-family object-detection model. The project will focus on a limited number of defect categories based on the availability and quality of the training dataset.

1.1 Problem Statement

Traditional fabric inspection often depends on manual observation. Inspectors must examine fabric surfaces carefully to identify visible defects, record findings, and prepare inspection reports. This process can be time-consuming, especially when many fabric pieces need to be checked.

Manual inspection may also produce inconsistent results because different inspectors may interpret small defects differently. Maintaining inspection records and reviewing previous findings can create additional work.

AI-based computer vision can help address these problems by analyzing fabric images and locating selected visible defects automatically. Therefore, FabricIQ AI aims to develop a web-based system that combines AI-assisted defect detection, image processing, human review, and digital inspection records.

1.2 Purpose

The primary objectives of FabricIQ AI are:

1. To develop a web-based fabric inspection application.
2. To implement an AI model that detects selected visible fabric defects.
3. To display detected defects with class labels, bounding boxes, and confidence scores where available.
4. To estimate dominant fabric colors using OpenCV.
5. To calculate a preliminary visual inspection score using documented rules.
6. To allow users to review and correct AI-generated findings.
7. To store inspection records and provide inspection history.
8. To generate structured inspection reports.
9. To identify low-confidence predictions and indicate when human review is required.

1.3 Scope

The system will do the following:

- Allow authorized users to log in.
- Allow users to upload fabric images or capture images through a supported camera.
- Detect selected visible fabric defects using a trained AI model.
- Display predicted defect classes, bounding boxes, and confidence scores where supported.
- Estimate dominant colors using OpenCV.
- Calculate a preliminary visual inspection score based on documented rules.
- Allow inspectors to confirm, reject, or correct AI predictions.
- Store inspection results and retrieve previous records.
- Generate inspection reports and display basic defect statistics.
- Warn users about low-confidence predictions and processing failures.

The system will not do the following:

- Detect every possible fabric defect or guarantee perfect predictions.
- Identify exact fiber composition from an ordinary image.
- Measure tensile strength, chemical properties, or colorfastness.
- Replace laboratory textile testing or professional inspection.
- Generate new fabric designs using generative AI.
- Perform design similarity search or live fashion-market prediction.
- Approve fabric for production automatically without human review.

The system's visual inspection score will be an estimate based on defined rules, not a certified textile-quality measurement. The AI model will support only the defect categories represented in its training data.

2. Design and Implementation Constraints

2.1 Programming Language

Python: Python will be used for AI model integration, image processing, model evaluation, and backend development. It provides suitable libraries for computer vision and deep learning.

JavaScript and React.js: React.js will be used to develop the web interface, including image upload, camera capture, result visualization, and dashboard pages.

CSS or Tailwind CSS: These technologies will be used to create a clear and responsive user interface.

Justification: This technology stack separates the frontend from the AI and backend components, making the application easier to develop and maintain.

2.2 Database & Servers

Database: MySQL

MySQL will store user accounts, inspection metadata, detected defects, reviewed results, and report information. Images will be stored as files or in suitable file storage, while their paths or references will be stored in the database.

Backend: FastAPI

FastAPI will provide REST API endpoints for image submission, AI inference, saving inspection results, retrieving inspection history, and generating reports. The React frontend will communicate with the backend through HTTP requests.

Image Processing: OpenCV

OpenCV will be used for image validation, resizing, camera-frame handling, and dominant-color estimation.

Version Control: Git and GitHub

Git and GitHub will be used for source-code management, collaboration, and tracking development progress.

Deployment: The system will first be developed and tested locally. Deployment to a suitable hosting service will be attempted if the available computing resources and hosting limits support the AI model.

2.3 AI Model & Services

Primary AI Model: YOLO-Family Object Detector

A suitable YOLO-family object-detection model will be used to identify selected visible fabric defects and estimate their locations in submitted images. The model will be implemented through a compatible PyTorch-based library.

The team will start with an existing model implementation and pretrained weights where appropriate. Fine-tuning will be attempted if a suitable labeled fabric-defect dataset is available and compatible with the selected model.

Dataset: The team will investigate public fabric-defect datasets, including KAGGLE and other potentially suitable datasets. Dataset availability, license, defect categories, annotations, and compatibility must be checked before selecting the final dataset.

Supporting Techniques:

- OpenCV will estimate dominant fabric colors.
- A rule-based function will calculate a preliminary visual inspection score using confirmed findings and documented severity rules.
- Model confidence thresholds will help flag uncertain predictions for human review.

Evaluation: The model will be evaluated using precision, recall, mAP where appropriate, inference time, and visual examination of successful and unsuccessful predictions.

Justification: Object detection is appropriate because the project needs to identify the locations of visible defects, not merely assign a label to the whole image. Using an existing model implementation reduces the need to build a deep-learning architecture from scratch. However, dataset preparation, training or fine-tuning, integration, and evaluation will still be required.

3. User Classes and Characteristics

Fabric Inspector

- Log in to the application.
- Upload fabric images or capture images through a supported camera.
- Run AI-assisted inspections.
- View predicted defects, bounding boxes, and confidence scores.
- Review and correct AI-generated findings.
- View the preliminary visual inspection score.
- Save inspection results and view permitted inspection history.

Inspection Manager

- View available inspection records.
- Review inspection summaries and basic defect statistics.
- Search or filter previous inspections.
- Generate inspection reports.
- Review confirmed inspection outcomes.

System Admin

- Manage user accounts and assigned roles.
- Maintain basic application settings.
- Manage supported system settings where implemented.
- Monitor application errors and maintain system data.

4. Requirement Elicitation

Requirement elicitation is an important phase of the Software Requirements Specification (SRS) process. It involves identifying stakeholder needs, understanding existing problems, collecting expectations, and translating the collected information into system requirements.

For FabricIQ AI, requirement elicitation focuses on understanding the challenges associated with fabric inspection, the information required in inspection reports, the usefulness of AI-based defect detection, and the importance of human review. The selected elicitation techniques are semi-structured interviews and a questionnaire. These techniques are appropriate for collecting both detailed opinions and structured feedback from people who have relevant experience with fabric inspection, sewing, garment production, or textile studies.

The project team also conducted a scope and technical-feasibility review to ensure that the proposed features could reasonably be attempted by a three-member academic team. The final requirements were organized according to their importance to the primary objective of the system: detecting selected visible fabric defects and managing inspection results.

4.1 Elicitation Techniques

A. Semi-Structured Interviews

Semi-structured interviews were selected as a requirement-elicitation technique because they allow the interviewer to ask consistent questions while also exploring relevant answers in more detail. The interview questions focus on current inspection practices, common defects, manual inspection difficulties, reporting needs, and expectations regarding AI-generated results.

The intended participants are people with relevant experience in fabric inspection, sewing, garment production, or textile studies. The following questions form the interview guide for collecting stakeholder requirements.

Interview Questions

1. How is fabric currently inspected to identify visible defects?
2. Which types of visible fabric defects are most important to identify?
3. What difficulties may occur during manual fabric inspection?
4. How are inspection findings currently recorded and retrieved?
5. Would highlighting the approximate location of a detected defect help an inspector? Why?
6. What information should be included in a fabric-inspection report?
7. Should an inspector be able to confirm, reject, or correct an AI-generated prediction before saving the result?
8. How should the system handle uncertain or potentially incorrect AI predictions?
9. Would inspection history and basic defect statistics be useful? Which information would be most valuable?
10. Which features should be prioritized in the first version of FabricIQ AI?

The interview guide is intended to identify functional requirements and quality expectations. Responses should be summarized without unnecessarily disclosing participants' personal information. The questions may be followed by clarification questions where needed.

B. Questionnaire

A questionnaire was selected as a complementary elicitation technique. It is intended to collect structured feedback about inspection difficulties, useful defect categories, desired report information, and the proposed system features.

The questionnaire can be distributed using Google Forms or another suitable method to available participants with relevant experience. The following questions form the questionnaire used for requirement collection.

Questionnaire Questions

Q1. What is your level of experience with fabric inspection or garment-related work?

- No direct experience
- Student or trainee
- Some practical experience
- Experienced worker or inspector
- Other

Q2. Which methods are used for fabric inspection in your experience?

- Manual visual inspection
- Digital or image-based inspection
- Both manual and digital methods
- Not sure
- Other

Q3. What difficulties can occur during manual fabric inspection? Select all that apply.

- Small defects may be missed.
- Inspection may require considerable time.
- Different inspectors may reach different conclusions.
- Recording and retrieving results may be difficult.
- Preparing reports may require additional work.
- Other.

Q4. Which visible defect categories would be useful for an AI-assisted system to identify? Select all that apply.

- Holes or tears
- Stains or spots
- Broken threads or yarn irregularities
- Slubs or thick yarn irregularities
- Other visible defects
- Not sure

The final supported categories will depend on the chosen dataset and its annotations.

Q5. How useful would it be to display a bounding box around a predicted defect?

- Very useful
- Useful
- Neutral
- Not very useful
- Not useful

Q6. What information should be included in an inspection report? Select all that apply.

- Fabric image
- Predicted defect categories
- Defect locations
- Confidence scores
- Inspector corrections or comments
- Preliminary visual inspection score
- Inspection date and reviewer
- Other

Q7. Should an inspector be able to confirm, reject, or correct AI predictions before saving the final inspection?

- Yes
- No
- Not sure

Q8. What should the system do when the AI model produces an uncertain prediction?

- Display a warning and request human review.
- Display the prediction without a warning.
- Hide the uncertain prediction.
- Other.

Q9. How useful would inspection history and basic defect statistics be?

- Very useful
- Useful
- Neutral
- Not very useful
- Not useful

Q10. Which features should be prioritized in the first version of FabricIQ AI? Select up to five.

- User login and role-based access
- Fabric image upload or camera capture
- AI-based defect detection
- Defect bounding boxes and confidence display
- Dominant-color estimation
- Preliminary visual inspection score
- Human review and correction
- Inspection history
- Inspection reports
- Basic statistics

- Other

Q11. What concerns might you have about using AI for fabric inspection? Select all that apply.

- Incorrect predictions
- Defects being missed
- Poor lighting or image quality affecting results
- Difficulty understanding the AI output
- Privacy or storage of uploaded images
- Other
- No major concerns

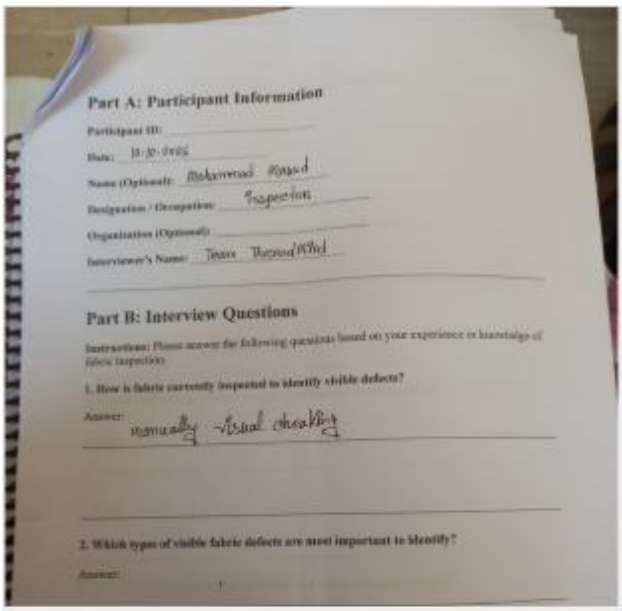
Q12. What additional suggestions do you have for improving an AI-assisted fabric-inspection system?

This question allows participants to provide open-ended suggestions that may identify needs not covered by the predefined options.

4.2 Elicitation Evidence

The following evidence was identified during requirements analysis and technical feasibility review.

ID	Evidence	Related Requirements
E-01	Fabric defects need to be detected and recorded.	FR-04, FR-09
E-02	Defects should be shown with labels and bounding boxes.	FR-05
E-03	Users need image upload and inspection reports.	FR-03, FR-10
E-04	AI predictions need human review and confidence warnings.	FR-08, FR-11
E-05	Inspection history and statistics support quality tracking.	FR-09, FR-12
E-06	Detection depends on the availability of suitable labeled data.	FR-04
E-07	Design recommendations, similarity search, trend prediction, care-label OCR, and fabric-type classification were excluded to keep the project feasible.	Scope Decision



Part A: Participant Information

Participant ID: _____

Date: 10-10-2025

Name (Optional): Mohammad Masud

Designation / Occupation: Inspection

Organization (Optional): _____

Interviewer's Name: Tasnia Tasnim Mim

Part B: Interview Questions

Instruction: Please answer the following questions based on your experience or knowledge of fabric inspection.

1. How is fabric currently inspected to identify visible defects?

Answer: manually - visual checking

2. Which types of visible fabric defects are most important to identify?

Answer: _____



Current detection process is time consuming and increase cost

Most important requested features:

automated fabric defect detection

Additional suggestions or comments:

_____

Interview completed by: _____

Date: 10/10/25



4.3 Key Findings

Based on the proposed elicitation topics and the technical feasibility review, the following illustrative findings describe the needs that FabricIQ AI is designed to address.

KF-01: Need for Efficient Defect Inspection

Manual fabric inspection can involve repeated visual checking and manual recording. FabricIQ AI therefore includes an AI-assisted defect-detection feature to help identify selected visible defects.

KF-02: Need for Clear Defect Visualization

A defect label alone may not show where the problem occurs. The system will display bounding boxes, predicted classes, and confidence scores where available so that users can examine the predicted locations.

KF-03: Need for Flexible Image Input

The system will support image upload and, where the browser and device permit, camera capture. This provides two practical ways to submit fabric images for inspection.

KF-04: Need for Human Review

AI predictions may include false positives or missed defects. The system will allow inspectors to confirm, reject, or correct predictions before saving the reviewed results.

KF-05: Need for Organized Inspection Records

A database-backed inspection history will allow authorized users to retrieve previous inspection records instead of relying exclusively on separate manual records.

KF-06: Need for Structured Reports

Inspection reports will summarize the submitted image reference, detected findings, review outcome, preliminary visual score, and inspection date.

KF-07: Need for Basic Supporting Analysis

Dominant-color estimation and basic defect statistics will provide additional information about inspected images and saved inspection records. These functions will remain limited to what can be implemented and tested reliably.

4.4 Requirement Changes After Elicitation

The following sample change log demonstrates how the requirements may be refined following user feedback and technical feasibility analysis. It is a proposed change log and must be reconciled with the team's actual initial requirements and collected evidence.

ID	Change	Before Elicitation	After Elicitation	Source / Reason
FR-07	Score limitations clarified	Provide an inspection score.	Provide a preliminary visual score based on documented rules and reviewed findings.	Technical feasibility and avoidance of unsupported quality claims.
FR-09	Inspection history retained	Save inspection results.	Allow authorized users to retrieve previous inspection records.	Proposed record-management improvement.
FR-10	Report content specified	Generate an inspection report.	Include findings, review outcome, score, image reference, and inspection date.	Proposed reporting improvement; confirm against actual feedback.
Scope-01	Design similarity search excluded	Consider design similarity search as an optional feature.	Exclude it from the initial version.	Project-scope and implementation-complexity review.
Scope-02	AI-Based Design Recommendation	The system was planned to recommend fabric designs based on analyzed fabric images.	Removed from the final scope.	Dataset feasibility review
Scope-03	Fashion Trend Prediction	The system was planned to analyze fashion trends and recommend trending fabric designs.	Removed from the final scope.	Dataset feasibility review
Scope-04	Care-Level Detection	The system was initially planned to identify fabric care instructions by reading care labels from images.	Removed from the final scope.	Technical feasibility review: reliable care-label text recognition would require suitable labeled examples, OCR integration, and additional validation.

Scope-05	Fabric Type Classification	The system was initially planned to classify fabric images into fabric types, such as cotton, silk, or polyester.	Removed from the final scope.	Dataset feasibility review

5. Functional Requirements (FR)

Functional requirements describe the main functions that the FabricIQ AI system must perform. The following requirements define the core features of the proposed system.

ID	Requirement Name	Description	Users	Priority
FR-01	User Login	The system shall allow authorized users to log in using valid credentials.	All authorized users	High
FR-02	Role-Based Access	The system shall restrict available functions according to the user's assigned role.	Inspector, Manager, Admin	Medium
FR-03	Image Input	The system shall allow users to upload a fabric image or capture an image using a supported camera.	Inspector	High
FR-04	AI Defect Detection	The system shall process a submitted image using the selected AI model and return predictions for supported defect categories.	Inspector	High
FR-05	Defect Visualization	The system shall display detected defects with bounding boxes, class labels, and confidence scores where available.	Inspector	High
FR-06	Color Estimation	The system shall estimate and display dominant fabric colors using image-processing methods.	Inspector	Medium
FR-07	Visual Inspection Score	The system shall calculate a preliminary visual score using documented rules based on detected or confirmed defects.	Inspector, Manager	Medium
FR-08	Human Review	The system shall allow an inspector to confirm, reject, or correct AI-generated findings before saving the final result.	Inspector	High
FR-09	Inspection History	The system shall store inspection metadata and allow authorized users to retrieve previous inspection records.	Inspector, Manager	High

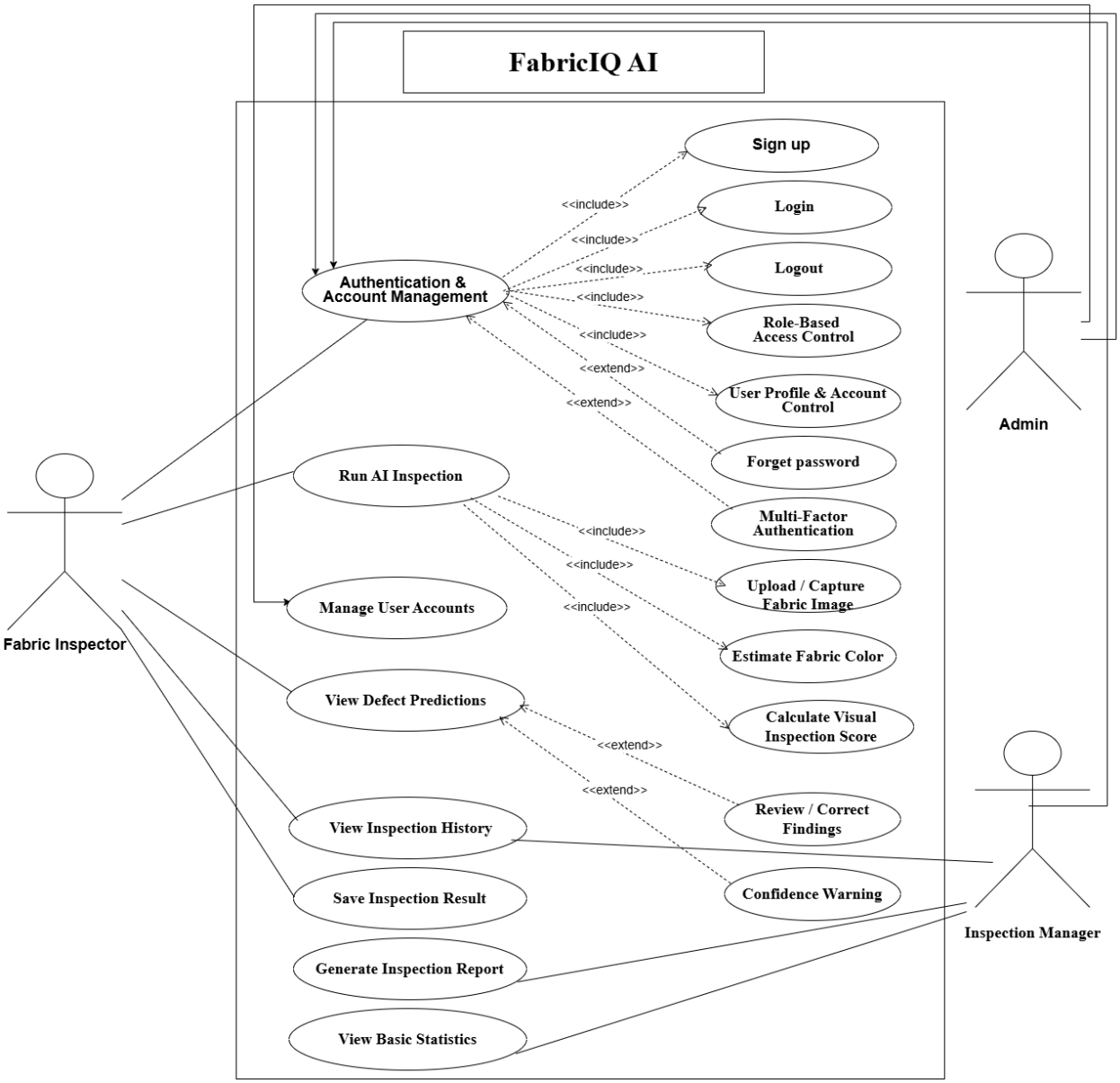
FR-10	Inspection Report	The system shall generate a structured report containing the image reference, findings, reviewed outcome, visual score, and inspection date.	Inspector, Manager	Medium
FR-11	Confidence Warning	The system shall identify predictions below a configured confidence threshold and indicate that human review is required.	Inspector	High
FR-12	Basic Statistics	The system shall summarize saved inspection records, including the number of inspections and counts of confirmed defect categories.	Manager	Medium

6. Non-Functional Requirements

1. **Performance:** The system should process fabric images within a reasonable time.
2. **Security:** User passwords must be securely stored, and access must be role-based.
3. **Usability:** The interface should be simple and user-friendly.
4. **Reliability:** The system should display clear error messages when processing fails.
5. **Compatibility:** The application should work on supported modern web browsers.
6. **Data Integrity:** The system must validate data before saving inspection records.
7. **AI Quality:** The model should be evaluated using precision, recall, and mAP where applicable.
8. **Transparency:** AI predictions should show confidence scores where available and support human review.
9. **Maintainability:** The code should be organized into separate frontend, backend, AI, and database modules.

7. System Design Diagrams

7.1 Use Case Diagram:



7.2 Use Case Descriptions

This section describes the main success flows of the high-priority functional requirements of FabricIQ AI. Each use case explains how the user interacts with the system to complete a specific task successfully.

Use Case 1: User Login

Use Case ID: UC-01
Related Requirement: FR-01
Priority: High
Primary Actor: Fabric Inspector, Inspection Manager, System Admin
Precondition: The user has a registered account with valid credentials.
Postcondition: The user is successfully authenticated and can access the application.

Main Success Flow:

1. The user opens the FabricIQ AI web application.
2. The system displays the login page.
3. The user enters their registered username or email and password.
4. The user submits the login form.
5. The system validates the submitted credentials.
6. The system authenticates the user.
7. The system opens the appropriate dashboard for the authenticated user.

Use Case 2: Upload or Capture Fabric Image

Use Case ID: UC-02
Related Requirement: FR-03
Priority: High
Primary Actor: Fabric Inspector
Precondition: The inspector is logged in to the system.
Postcondition: A valid fabric image is available for AI-assisted inspection.

Main Success Flow:

1. The Fabric Inspector opens the fabric inspection interface.
2. The system displays the image upload and camera capture options.
3. The inspector selects an image from the device or captures an image using a supported camera.
4. The system receives the selected or captured image.
5. The system checks that the image is in a supported format and can be processed.
6. The system displays an image preview to the inspector.
7. The inspector confirms the image for inspection.
8. The system makes the image available for the AI inspection process.

Use Case 3: AI-Based Fabric Defect Detection

Use Case ID: UC-03
Related Requirement: FR-04
Priority: High
Primary Actor: Fabric Inspector
Precondition: The inspector is logged in, and a valid fabric image has been submitted.
Postcondition: The system produces defect predictions for the supported defect categories, where detection is successful.

Main Success Flow:

1. The Fabric Inspector submits a fabric image for inspection.
2. The system prepares the image for processing.
3. The system sends the image to the selected AI defect-detection model.
4. The AI model analyzes the image.
5. The model identifies possible defects belonging to its supported categories.
6. The model returns the predicted defect classes, locations, and confidence scores where available.
7. The system receives and processes the prediction results.
8. The system makes the results available for visualization and human review.

Use Case 4: Defect Visualization

Use Case ID: UC-04
Related Requirement: FR-05
Priority: High
Primary Actor: Fabric Inspector
Precondition: The AI inspection process has returned prediction results.
Postcondition: The inspector can examine the predicted defects and their locations in the fabric image.

Main Success Flow:

1. The Fabric Inspector opens the AI inspection results.
2. The system displays the inspected fabric image.
3. The system overlays bounding boxes around the detected defect locations where available.
4. The system displays the predicted defect class labels.
5. The system displays confidence scores where available.
6. The inspector examines the displayed predictions.
7. The inspector proceeds to human review if confirmation or correction is needed.

Use Case 5: Inspection History

Use Case ID: UC-05
Related Requirement: FR-09
Priority: High
Primary Actor: Fabric Inspector, Inspection Manager
Precondition: The user is logged in and has permission to access inspection records.
Postcondition: The system displays the inspection records available to the user.

Main Success Flow:

1. The user logs in to the FabricIQ AI system.
2. The user opens the Inspection History section.
3. The system retrieves the inspection records that the user is authorized to access.
4. The system displays the available records with relevant information, such as inspection date, image reference, and inspection outcome.
5. The user browses the inspection history.
6. The user selects a particular inspection record.
7. The system displays the details of the selected inspection, including saved defect findings and review results where available.

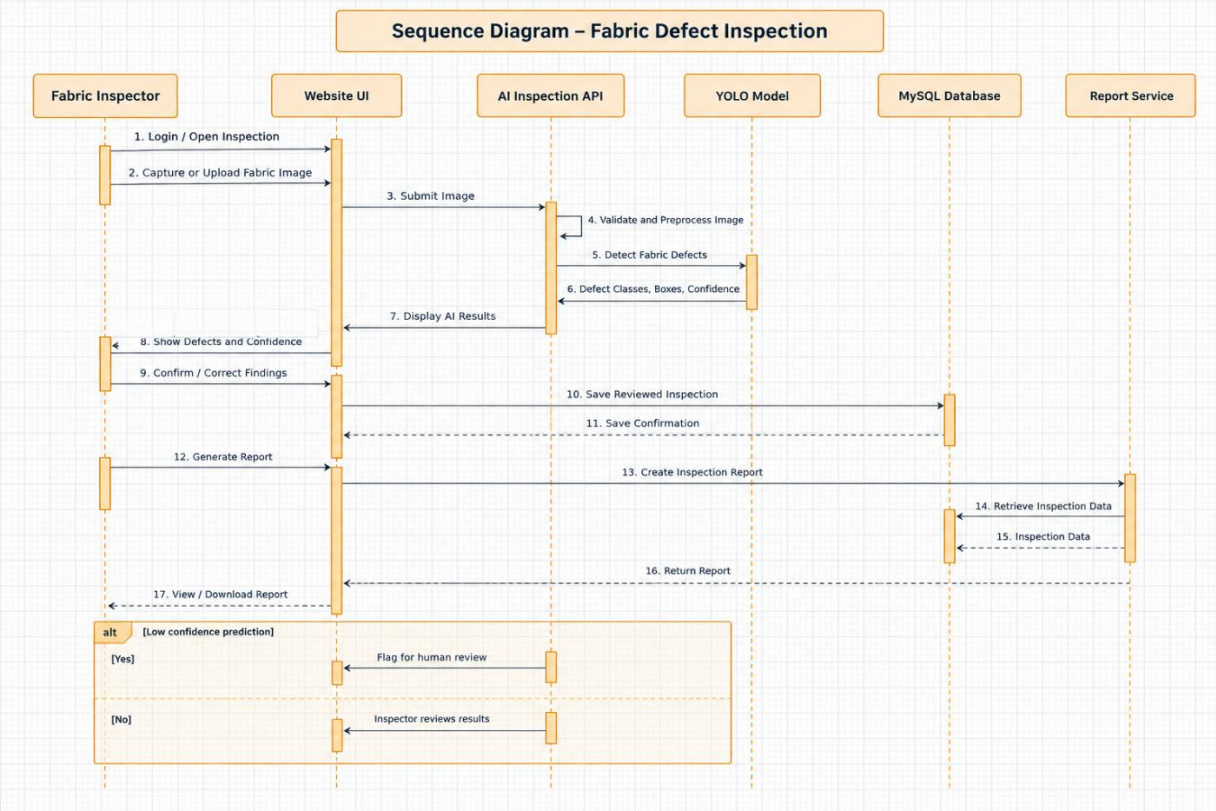
Use Case 6: Confidence Warning

Use Case ID: UC-06
Related Requirement: FR-11
Priority: High
Primary Actor: Fabric Inspector
Precondition: The inspector is logged in, and the AI model has returned defect predictions with confidence scores.
Postcondition: The inspector is informed about low-confidence predictions and can review the uncertain results.

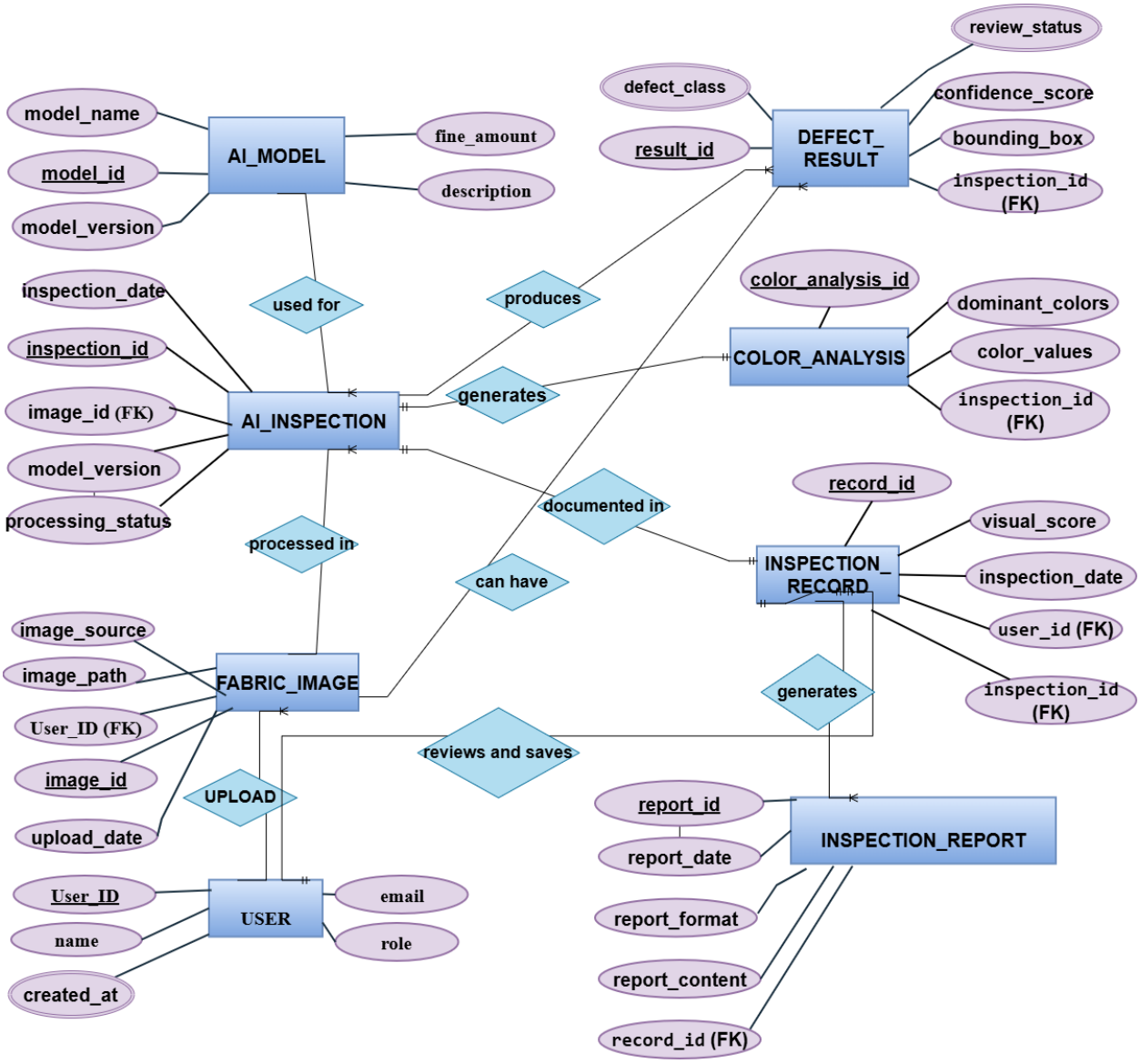
Main Success Flow:

1. The Fabric Inspector submits a fabric image for AI inspection.
2. The system processes the image using the selected AI defect-detection model.
3. The AI model returns defect predictions with confidence scores where available.
4. The system compares the confidence scores against the configured confidence threshold.
5. The system identifies predictions whose confidence scores fall below the threshold.
6. The system displays a warning to indicate that some predictions are uncertain.
7. The system highlights or identifies the uncertain predictions for the inspector.
8. The inspector reviews the uncertain predictions before confirming or saving the inspection results.

7.3 Activity Diagram:



7.4 Entity Relationship Diagram (ERD):

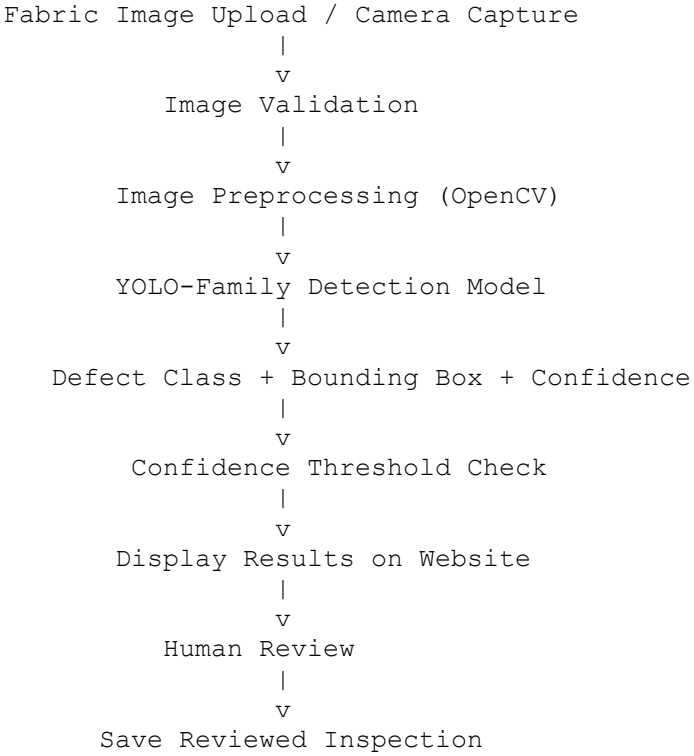


7.5 AI Feature Design

Data: The system will use fabric images containing visible defects from a suitable publicly available dataset. The team will investigate datasets such as TILDA and other fabric-defect datasets. The final dataset will be selected based on availability, licensing, image quality, defect categories, and annotation format. The number of images and supported defect classes will be documented after the dataset is obtained. The images and annotations will be checked and prepared for training, validation, and testing. If necessary, image resizing and other suitable preprocessing techniques will be applied.

Model: A YOLO-family object-detection model implemented using a compatible PyTorch-based library will be used to detect selected visible fabric defects. Object detection is suitable because it identifies both the defect category and its location in an image. The team will initially use an existing model implementation and pretrained weights where appropriate, reducing the need to build a deep-learning architecture from scratch. A pretrained model and the fine-tuned model will be compared where the dataset and available computing resources permit.

Pipeline:



Integration: The React.js frontend will send the fabric image to a FastAPI backend endpoint. The backend will validate the image and pass it to the YOLO model for inference. The model's predictions will be returned to the backend and displayed on the website. OpenCV will estimate dominant colors, while a rule-based function will calculate a preliminary visual inspection score.

After human review, the system will store the inspection details, defect predictions, review status, and related information in MySQL.

Evaluation & Fallback: The model will be evaluated using a separate test dataset. Evaluation metrics will include precision, recall, mean Average Precision (mAP), where appropriate, and inference time. The team will also examine false detections and missed defects. Actual performance results will be reported after testing rather than assumed in advance. Predictions below a configured confidence threshold will be flagged for human review. If the AI model is unavailable or an image cannot be processed, the system will display an error message and allow the user to retry. A failed prediction will never be treated as confirmation that the fabric is defect-free.

8. Team Member Responsibility Matrix

Member Name	ID	Primary Responsibilities
Md.Sajjad Hossain Nahid	241-35-241	Frontend UI/UX development using React.js, image upload and camera interface, inspection-result visualization, and dashboard design.
MD. Yasin	241-35-307	AI model integration, dataset investigation and preparation, YOLO-family model setup or fine-tuning, and model evaluation.
Sumaiya Akter Sumi	241-35-218	Backend API development using FastAPI, MySQL database design, authentication, inspection-history management, report generation, and backend integration.

All team members will participate in system integration, testing, debugging, documentation, and final project presentation. Each member will maintain their assigned responsibilities and coordinate with the others to ensure that the frontend, backend, database, and AI components work together.